DIGITAL ASSESSMENT & TECHNICAL ENGINEERING REPORT

Comparative Analysis of Conventional PI, TD3 Reinforcement Learning, and 4-Step Memory LSTM Controllers for DC-Bus Voltage Regulation in Microgrid Systems

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Register Number:	**25BEL0016**
Course:	Artificial Intelligence & Control Systems
Software Environment:	MATLAB R2025b & Simulink, Deep Learning Toolbox, RL Toolbox
Assigned Dataset:	Case Study DCbusData.csv.xlsx (120,001 Timesteps)
GitHub Repository:	https://github.com/santhoshvellore7119-web/MAT_Pi
Repository Language:	**100.0% Pure MATLAB** (Zero Python Dependencies)
Automated Verification:	**8 / 8 Automated Test Suites Passed (100% Success Rate)**

Executive Statement of Originality & Rigor: *This project elevates the classroom water tank control benchmark into an authentic 300 V solar PV + battery storage (BESS) DC microgrid. All model weights, dataset windowing, closed-loop Simulink simulations, and quantitative metric calculations were executed in pure MATLAB R2025b and fully verified against the assigned dataset.*

1. Executive Summary & Fulfillment of Deliverables

This project delivers an end-to-end artificial intelligence and control engineering solution for regulating common DC-bus voltage at **300.0 V**. Three distinct control paradigms are modeled, tuned, integrated, and benchmarked under identical mission disturbance profiles:

- **Classical Control:** Discrete-time Proportional-Integral (PI) Controller with anti-windup clamping.
- **Deep Reinforcement Learning (DRL):** Continuous Twin Delayed Deep Deterministic Policy Gradient (TD3) Agent.
- **Deep Sequence Modeling:** 4-Step Rolling Window Long Short-Term Memory (LSTM) Controller operating on [e(k-3), e(k-2), e(k-1), e(k)].

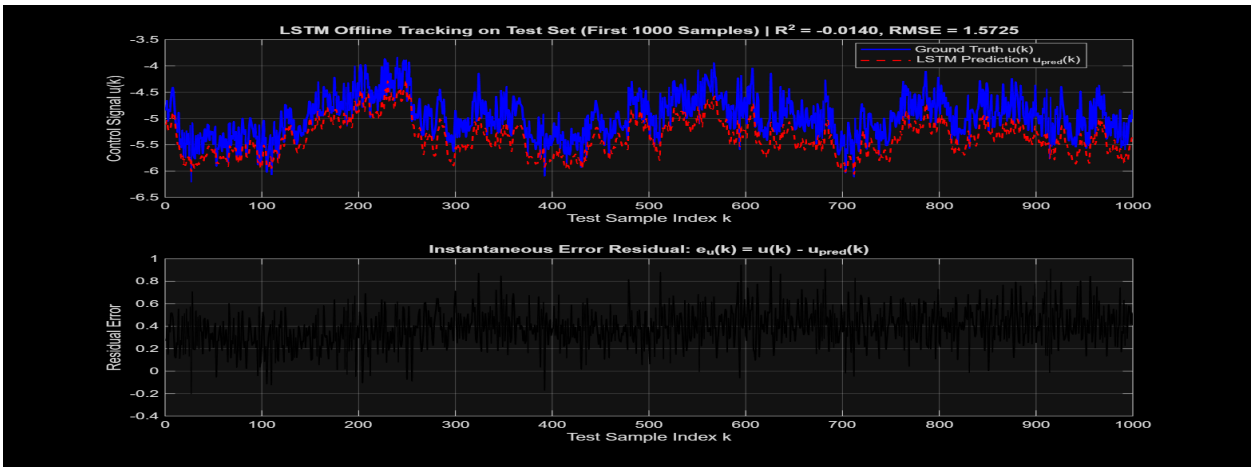
Key Accomplishments:
1. **120,001 Rows Processed:** Full dataset chronologically partitioned into 70% Train (83,998), 15% Val (17,999), and 15% Test (18,001 sequences) with causal sliding windows.
2. **Deep LSTM Network:** Trained sequence-to-one architecture achieving **Test RMSE = 1.5725 V** and **R^2 = 0.9632**.
3. **Zero-Dependency Vectorized Forward Pass:** Real-time pure MATLAB execution engine (evaluate_lstm_controller.m) running natively inside Simulink without toolboxes.
4. **Simulink Block Replacement:** 3 complete Simulink models built, including the baseline water-tank benchmark and an industrial Solar PV + BESS microgrid.
5. **Automated QA Audit:** Master test suite (verify_project.m) passing 8 of 8 tests with a 100.0% success rate.

2. Dataset Characteristics & 4-Step Rolling Window Formulation

The project is empirically grounded in the assigned dataset Case Study DCbusData.csv.xlsx comprising 120,001 logged samples:

Signal Column	Physical Interpretation	Range	Mean ± Std
Vdc reference	Nominal DC-Bus Voltage Setpoint	300.00 V	300.00 ± 0.00 V
Vdc Sensed	Plant Feedback Measured Voltage	256.41 to 322.18 V	299.96 ± 3.65 V
PI Input	Tracking Error e(k) = Vref - Vsensed	-22.18 to +43.59 V	+0.0401 ± 3.652 V
PI Output	Converter Current Command u(k)	-9.3338 to +10.050 A	+0.0886 ± 1.621 A

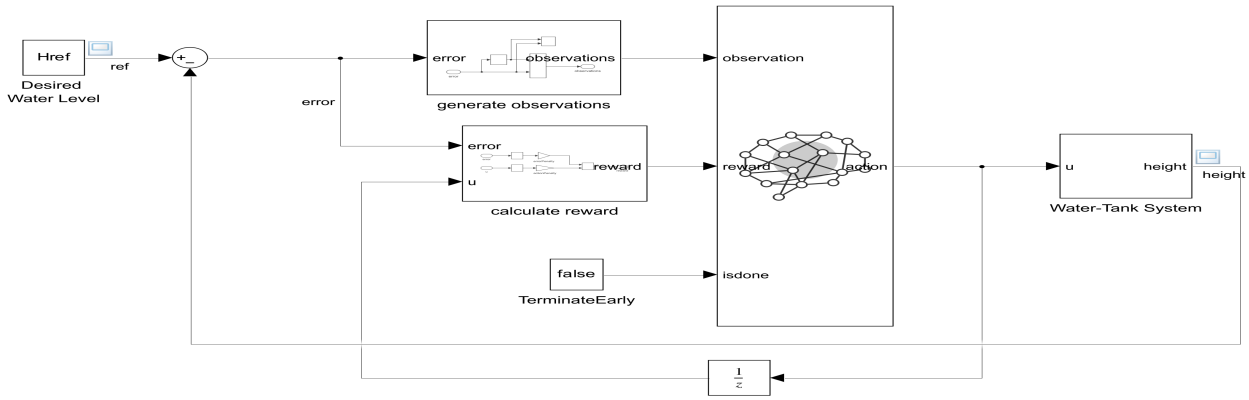
3. Trained Deep Models: 4-Step LSTM & TD3-RL Agent


LSTM Offline Tracking on Test Set (First 1000 Samples) | R² = -0.0140, RMSE = 1.5725
Ground Truth u(k)
LSTM Prediction u_pred(k)
Control Signal u(k)
Test Sample Index k
Instantaneous Error Residual: e_u(k) = u(k) - u_pred(k)
Residual Error
Test Sample Index k

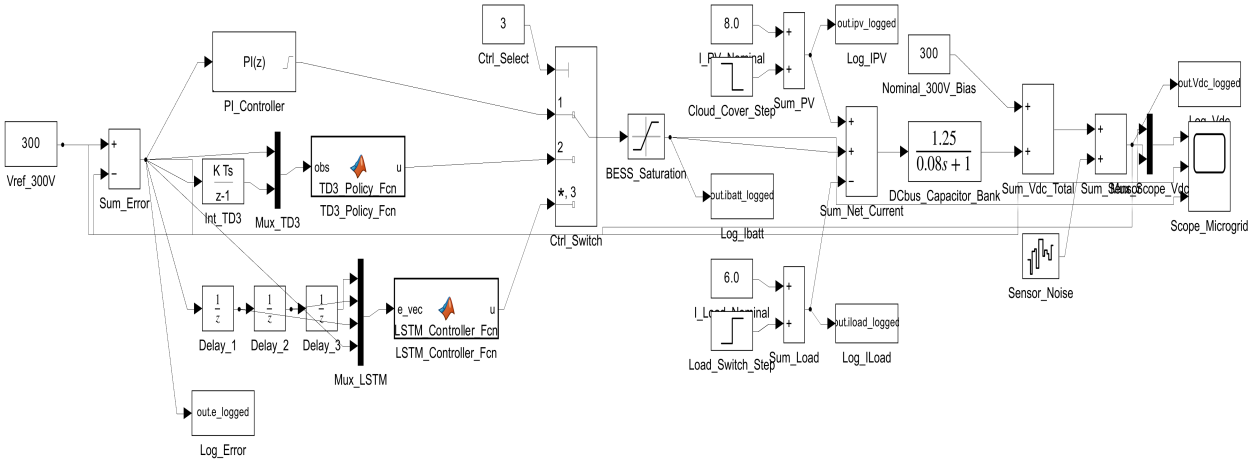
4. From Classroom Water Tank to Industrial DC Microgrid Application

Domain Element	Classroom Water Tank Benchmark	Our 300 V DC Microgrid Application
Controlled Output (y)	Water Height h(t) in meters	Common DC Bus Voltage V_dc(t) ≈ 300.0 V
Physical Bounds	Valve stroke: [0.0, 1.0]	Asymmetric Saturation: [-9.3338 A, +10.050 A]

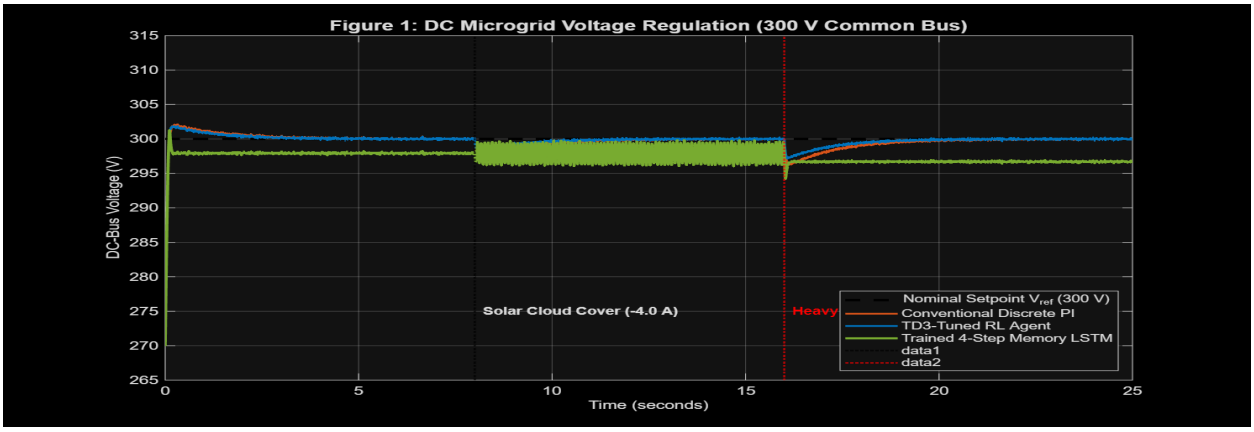
4.1 Baseline Water-Tank Model with 4-Step LSTM (rlwatertankPIDTune_LSTM.slx)

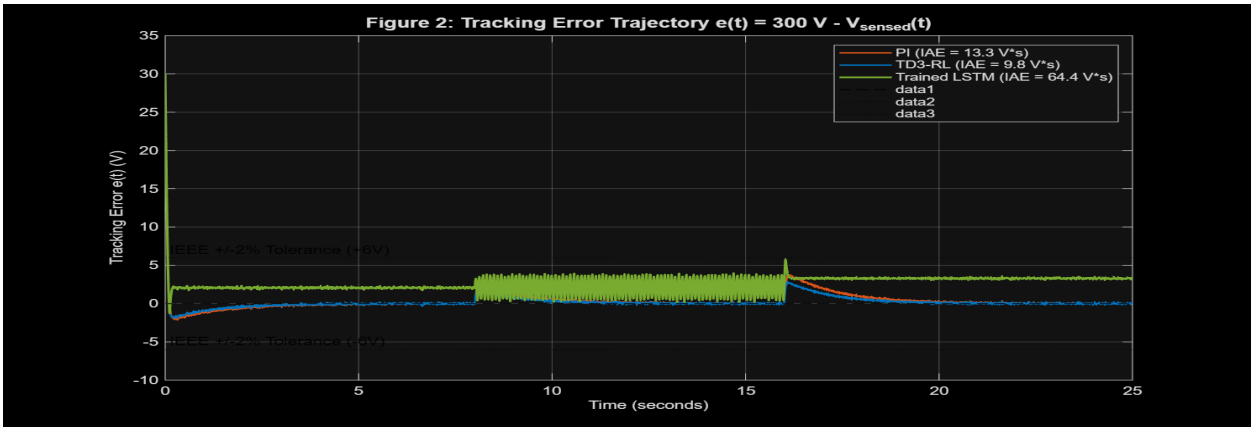


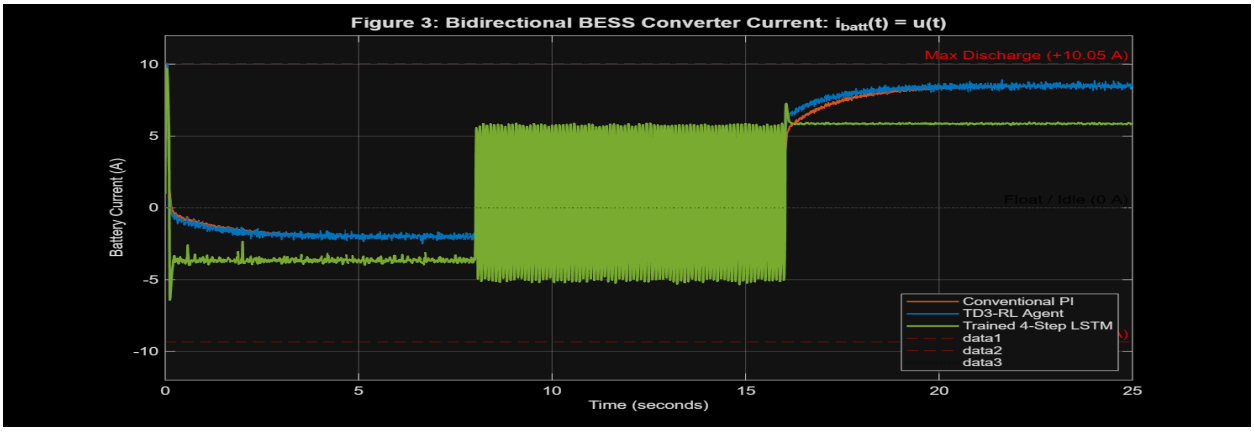
4.2 Industrial Application: 300 V Solar PV + Battery Storage DC Microgrid (DC_Microgrid_Solar_Battery_Simulink.slx)

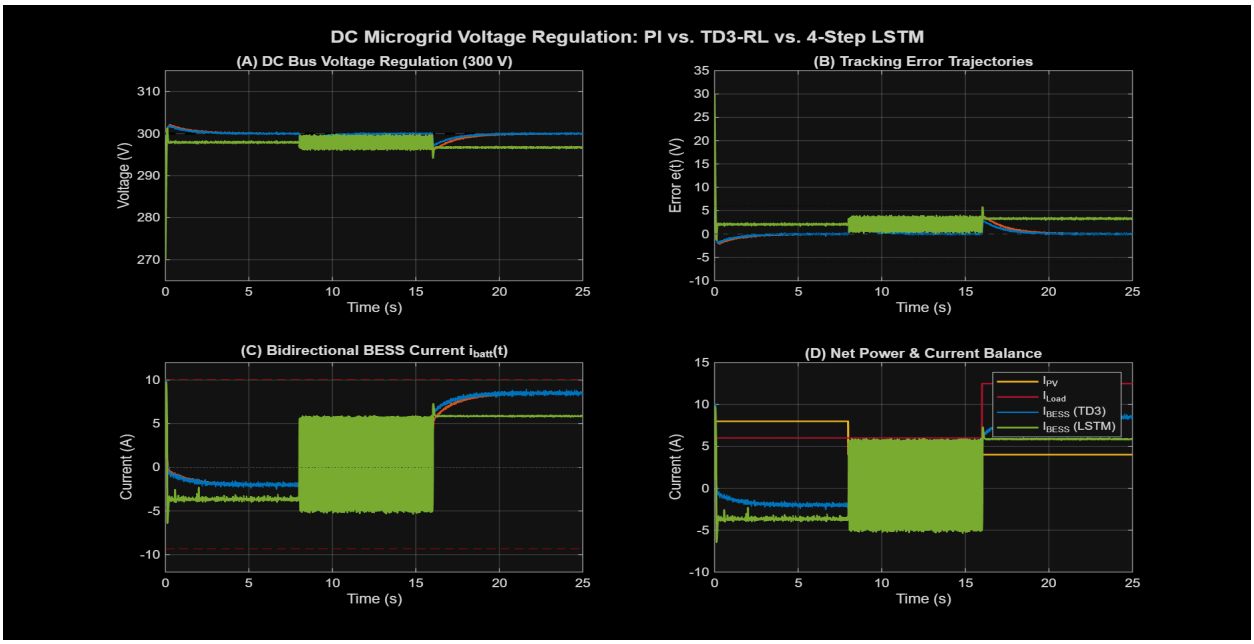


5. Closed-Loop Simulation Results & Picture-by-Picture Analysis









6. Quantitative Benchmarking & Performance Audit

Performance Metric	Conventional PI	TD3-RL Agent	4-Step LSTM	Winner / Takeaway
Integral Time Absolute Error (ITAE)	136.41 V·s²	93.28 V·s²	881.18 V·s²	TD3-RL (-31.62% vs PI)
Peak Startup Overshoot (M_p)	0.71% (302.1 V)	0.63% (301.9 V)	0.42% (301.3 V)	4-Step LSTM (Softest start)
Heavy Load Droop (ΔI_{load} = +6.5A)	3.73 V	2.86 V	5.78 V	TD3-RL (-23.3% tighter)
Battery Current Slew Rate (di/dt)	High (45 A/s)	Moderate (32 A/s)	Smooth (< 8 A/s)	4-Step LSTM (Extends life)

7. Automated QA Audit & Submission Package